

Application Architecture & Dependencies

1. System Overview

The system is a modular .NET 8 suite designed for automated requirement validation. It translates static specifications into dynamic network-verified test cases.

Presentation Layer: TestCreatorWpfApp

The central command hub. Manages UI, user history, persistent settings, and controls external terminal lifecycles.

Logic Engine: TestCreator

Parses complex boolean logic (AND/OR/XOR/NOT) from Word documents and generates high-volume test permutations.

Communication Layer: ConnectionManager

Implements robust TCP and UDP protocols for real-time echo verification and SQL Server integration.

2. External Dependencies

- `DocumentFormat.OpenXml` - Microsoft Office Integration
- `Microsoft.Data.SqlClient` - SQL Server Data Provider
- `System.Text.Json` - Lightweight persistence & configuration

3. Data Flow

Word Requirements -> Logic Parser -> Permutation Generator -> Excel Export -> Network Client (TCP/UDP) -> Remote Server -> Result Verification -> UI Dashboard.